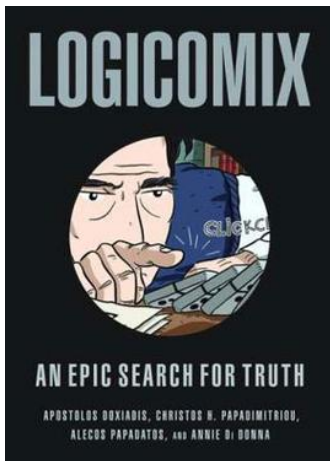


Logic & the search for truth

www.logicomix.com



Challenges to

- truth

Liar's Paradox: 'I am lying'

Challenges to

- truth

Liar's Paradox: 'I am lying'



P: *Trump is lying.*

T: *Putin is telling
the truth.*

Challenges to

- truth

Liar's Paradox: 'I am lying'

- sets (membership $\in$)

Russell set $R = \{x \mid \text{not } x \in x\}$

Challenges to

- truth

Liar's Paradox: 'I am lying'

- sets (membership $\in$)

Russell set $R = \{x \mid \text{not } x \in x\}$

$$R \in R \iff \text{not } R \in R$$

Challenges to

- truth

Liar's Paradox: 'I am lying'

- sets (membership $\in$)

Russell set $R = \{x \mid \text{not } x \in x\}$

- search (one by one)

Cantor: subsets of $\{0, 1, 2, \dots\}$ are uncountable

s_1	=	0	0	0	0	0	0	0	0	0	0	0	0	...
s_2	=	1	1	1	1	1	1	1	1	1	1	1	1	...
s_3	=	0	1	0	1	0	1	0	1	0	1	0	...	
s_4	=	1	0	1	0	1	0	1	0	1	0	1	...	
s_5	=	1	1	0	1	0	1	1	0	1	0	1	...	
s_6	=	0	0	1	1	0	1	1	0	1	1	0	...	
s_7	=	1	0	0	0	1	0	0	1	0	0	...		
s_8	=	0	0	1	1	0	0	1	1	0	0	1	...	
s_9	=	1	1	0	0	1	1	0	0	1	1	0	...	
s_{10}	=	1	1	0	1	1	1	0	0	1	0	1	...	
s_{11}	=	1	1	0	1	0	1	0	0	1	0	0	...	
$\vdots$		$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	

$$C = \{n \mid \text{not } n \in s_n\}$$

$$s = 10111010011...$$

Challenges to

- truth
Liar's Paradox: 'I am lying'
- sets (membership $\in$)
Russell set $R = \{x \mid \text{not } x \in x\}$
- search (one by one)
Cantor: subsets of $\{0, 1, 2, \dots\}$ are uncountable
- deduction
Gödel: consistency is not provable

Challenges to

- truth
Liar's Paradox: 'I am lying'
- sets (membership $\in$)
Russell set $R = \{x \mid \text{not } x \in x\}$
- search (one by one)
Cantor: subsets of $\{0, 1, 2, \dots\}$ are uncountable
- deduction
Gödel: consistency is not provable
- computation
Turing: Halting Problem is uncomputable

Challenges to

- truth

Liar's Paradox: 'I am lying'

- sets (membership $\in$)

Russell set $R = \{x \mid \text{not } x \in x\}$

- search (one by one)

Cantor: subsets of $\{0, 1, 2, \dots\}$ are uncountable

- deduction

Gödel: consistency is not provable

- computation

Turing: Halting Problem is uncomputable

Given a program P and data D , return either 0 or 1 depending on whether P halts on input D

$$\text{HP}(P, D) := \begin{cases} 1 & \text{if } P \text{ halts on } D \\ 0 & \text{otherwise.} \end{cases}$$

Challenges to

- truth

Liar's Paradox: 'I am lying'

- sets (membership $\in$)

Russell set $R = \{x \mid \text{not } x \in x\}$

- search (one by one)

Cantor: subsets of $\{0, 1, 2, \dots\}$ are uncountable

- deduction

Gödel: consistency is not provable

- computation

Turing: Halting Problem is uncomputable

- ▶ Turing test $\approx$ *can computers fool humans?* (CSU33061 AI 1)

Challenges to

- truth

Liar's Paradox: 'I am lying'

- sets (membership $\in$)

Russell set $R = \{x \mid \text{not } x \in x\}$

- search (one by one)

Cantor: subsets of $\{0, 1, 2, \dots\}$ are uncountable

- deduction

Gödel: consistency is not provable

- computation

Turing: Halting Problem is uncomputable

- ▶ Turing test $\approx$ *can computers fool humans?* (CSU33061 AI 1)
- ▶ Logic programming: *program via logic* (CSU34011)